

Thomas McMaster BSc (Hons) **Data Engineer / SQL Developer**

Contact information:

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Personal statement:

Experienced Data Engineer and SQL Developer, specializing in Synapse Analytics, PySpark, Spark SQL, SQL Server, and Kimball data modeling. Proven track record implementing medallion architectures, scalable 140+ entity semantic models, and transforming complex operational data by automating data pipelines to produce analytical data sets ready for business insights.

Professional experience:

August 2024 - Present

Data Engineer - ACCA

Technologies: Azure Resources, Synapse Analytics, PySpark, Spark SQL, SQL Server, Azure DevOps, YAML, Kimball, Data Modelling, GitHub, Jira, Confluence, Oracle SQL Developer

- Collaborated within a 4-person engineering team, along with external consultants
- Introduced a semantic reporting model (140+ entities) serving data analysts scientists, and users
- Delivered a medallion architecture to create a stable base for our business reporting platform
- Transformed manual flat file extracts and manipulation into automated scheduled data pipelines
- Modelled data into Kimball dimension / fact tables, moving it from operational to analytical states
- Introduced Power BI dashboards for operational monitoring of data pipeline performance
- Worked closely with consultants, suppliers, and external experts on large-scale projects
- Managed service incidents and support tasks to remove bugs and ensure continuous improvement

March 2022 - August 2024

Data Engineer - Hymans Robertson

Technologies: SQL Server 2008 R2 to 2019, PowerShell, SSIS, SSRS, Azure Dev Ops, Synapse Analytics, Kimball Data Warehousing Concepts, YAML, Git

- Delivery of changes to pension scheme data with T-SQL scripting and stored procedures
- Introduction of a reporting data warehouse, moving on-premises data to Azure
- Automated anonymization of data between databases with PowerShell and YAML pipelines
- Developed bespoke risk transfer reporting for in-house actuaries and external insurance firms
- Data analysis to inform colleagues involved in the UK's national Pensions Dashboard initiative
- Provisioned Cloud resources via CI and CD YAML-based pipelines
- Collaboration with colleagues via Git and Azure Dev Ops (ADO)

September 2020 –March 2022**Senior Data and Integration Officer – East Renfrewshire Council.**

Technologies: SQL Server 2008 R2 to 2019, Oracle 12 to 19c, PowerShell, SSIS, SSRS, Git, RMAN, Power BI

- DBA with responsibility for the Authority's SQL Server and Oracle database estate
- Automation of data transfers with scripted use of WinSCP
- Delivery of new Power BI reporting suite to monitor the health of the Council's database estate
- Working with suppliers and third parties to ensure smooth data migrations to Cloud hosts
- Developed a nightly Oracle RMAN 'hot' backup solution to increase database availability
- Creation of SQL Server corruption checker, checking 450 SQL databases for issues each night
- Maintaining software including Civica Icon, Servitor, NEC's Revenue's and Benefits and Ohms.
- Ongoing review of database security, roles, and the release of encryption-at-rest for SQL Server

February 2019 – September 2020**Database Developer and BI Analyst – Maggie's.**

Technologies: SQL Server (2008 R2 to 2017), thankQ CRM, SSIS, SSRS, VBA, Microsoft Office

- Automated data pipelines (fundraising financials, external communications, event promotion)
- Created bespoke data extracts for fundraisers and campaign organisers
- Resolution of database performance issues, backup strategy and scheduled maintenance
- Requirements analysis with support staff, colleagues at 26 NHS hospitals, and stakeholders
- Processing of requests sent to the Database Service Team's inbox

June 2011 – February 2019**Systems Analyst / Programmer / Team DBA – Hymans Robertson**

Technologies: SQL Server 2008 R2 to 2016, T-SQL, SSIS, MS Access, VBA, SSRS, Civica UPM2.

- KPI and SLA reporting to Management
- Support for the UPM2 pension management system and its underlying SQL Server databases
- Troubleshooting / root cause analysis of support requests submitted by end users
- Stakeholder communication to identify issues, gather requirements and produce specifications
- Business process and workflow analysis
- Team DBA, releasing production changes, maintaining SQL instances, and applying third-party upgrades

Earlier professional experience:**June 2004 – June 2011**

Roles including Database Programmer (Contract) at Hewlett Packard, Asset / Systems Support Officer at East Ayrshire Council Roads and Transportation Service, I.T. Technician at Booth Welsh Automation Ltd, and Systems Analyst (Contract) at GlaxoSmithKline.

Technologies: SQL Server, SSRS, SSIS, Microsoft Office with VBA, Microsoft Office, Classic ASP, AspenTech processing engineering tools

Education:**1999-2003:**

- Honours Degree (2:1) in Computer Studies from Glasgow Caledonian University.

Professional qualifications and further learning:

- Microsoft Certified in Azure Fundamentals (AZ-900)
- Microsoft Certified in Azure Data Fundamentals (DP-900)